Computer Science Department - University of Batna 2
2INF - Information Systems
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Tutorial 1: The Company and its Information System

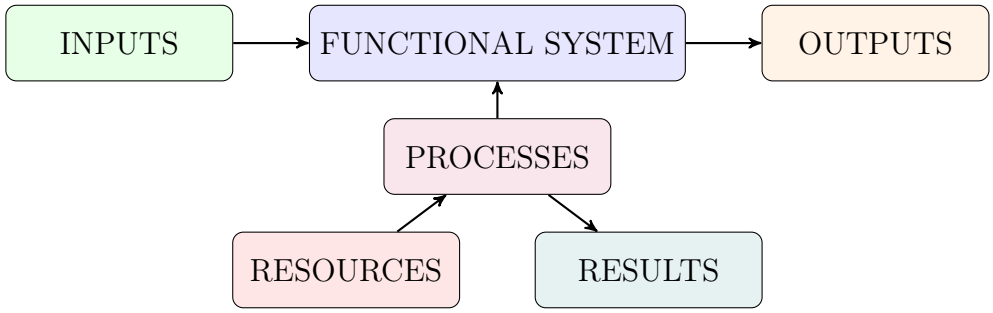
Exercise 1:

The Faculty of Mathematics and Computer Science at University of Batna 2 is organized into several interdependent entities. The objective of this exercise is to analyze this organization by identifying its functional systems and their interactions.

Definition to know

A **functional system** is an organized set of resources and processes that work together to achieve a specific function within the organization.

Functional System Diagram



Organization Description

Management

- **Dean** : responsible for overall management

Services and Departments

1. **General Secretariat** (4 services) :
    - Resources and Maintenance
    - Scientific, Cultural and Sports Activities
    - Budget and Accounting
    - Human Resources

2. **Scientific Council**
3. **Academic Departments** (3) :
    - Common Core
    - Computer Science
    - Mathematics
4. **Vice-Deans** (2) :
    - **A - Post-graduation and Research** :
        - Research Activities
        - Cooperation and External Relations
        - Post-Graduation Monitoring
    - **B - Studies and Students** :
        - Statistics, Information and Orientation
        - Teaching and Evaluation
        - Academic Affairs
5. **Central Library**

Questions

1. System Identification

List the 5 main functional systems of the faculty. For each system, specify:

- Its main mission
- The concerned services
- An example of input
- An example of output

2. Organizational Chart

Draw a simplified organizational chart of the faculty showing the main hierarchy and reporting lines.

3. Flow Analysis

For each document below, complete the table:

Document	**Concerned System**	**Input/Output**	**Processing Service**
Registration application			
Student certificate			
Student ID card			

4. Institution's Environment

Identify and characterize the external environment of the faculty.

Institution's Environment

Based on the organizational context, identify the external actors with which the faculty interacts. Classify these actors according to the following categories and specify for each the nature of the relationships:

- **Oversight and regulatory authorities**
- **Academic and scientific partners**
- **Socio-economic actors**
- **Users and beneficiaries**
- **Suppliers and service providers**

Expected format: Synthetic table or stakeholder mapping.

5. Information Systems (IS), Management Systems (MS) and Operational Systems (OS)

Distinguish and characterize the different types of systems present in the institution.

Information Systems (IS)

For each category of systems below, provide at least two concrete examples specific to the faculty context, specifying their purpose:

- **IS (Information Systems)** : Systems that manage information flows
- **MS (Management Systems)** : Systems that assist in strategic decision-making
- **OS (Operational Systems)** : Systems that execute core activities

Expected format: Three-column table with justified examples.

Information Systems (IS)	**Operational Systems (OS)**	**Management Systems (MS)**
Example 1:	Example 1:	Example 1:
Example 2:	Example 2:	Example 2:

Exercise 2: Grade Management System Analysis

Grade Management System Context

The Faculty of Mathematics and Computer Science wishes to modernize its grade management system. Currently, the process is semi-manual:

- Teachers fill out paper evaluation forms
- The academic affairs service manually enters grades into a spreadsheet
- Grade reports are printed and physically distributed
- Students must come in person to consult their results

Analyze this system and propose an integrated IT solution.

Questions

1. Process Analysis

Identify and describe the main processes of the current grade management system:

Process	**Concerned Actors**	**Documents Used**	**Identified Problems**
Grade entry			
Result validation			
Grade publication			
Document archiving			

2. Flow Modeling

Represent the information flow for managing an exam grade, from its entry to its publication.

Elements to include: Teacher → Evaluation form → Academic affairs service → Database → Student

3. New System Specification

Propose functional specifications for the new grade management system:

Required Main Functionalities

- Online grade entry by teachers
- Automatic grade range validation
- Automatic average calculation
- Secure result publication
- Complaint management

Actor and permissions table:

Actor	Permissions	Constraints
Teacher		
Program coordinator		
Academic affairs service		
Student		

4. Simplified Technical Architecture

Following the analysis of the current system, design the information flow diagram for the optimized grade management system. Your design should include the following technical components:

Components to consider

- User interface (Web)
- Application server
- Database
- Authentication system

Describe precisely the journey of an exam grade from its entry by the teacher to its consultation by the student.

5. Advantages of the New System

List the main advantages of the new system compared to the old one:

For teachers	For students	For administration